

OlistBR Delivery Analytics

Uncovering the Root Causes of **Late Deliveries** using Microsoft Fabric

Presented by the DAH-Namic Four
Junior Data Engineer Programme (Phase 1, 2 & 3)





DAIH-Namic Four

Junior Data Engineer
Programme
(Phase 1, 2 & 3)



Massive Scale

100,000+ orders across
all 27 Brazilian states,
capturing 2016–2018
e-commerce logistics data.



The Objective

Transform 9 raw Kaggle
CSV files (50+ MB) into a
prescriptive, actionable
logistics strategy.



The Core Problem: A 6.65% late delivery rate, averaging 10.64 days of delay.

Customers are waiting. Reputations are at risk. The business lacks visibility into why.

6.65% LATE DELIVERY DELIVERY RATE

Averaging **10.64 days of delay**
across the affected orders.



Does **physical distance** dictate the delay?



Does **premium freight pricing** guarantee speed?



Does **parcel weight** break the logistics chain?



Customers are waiting. Reputations are at risk. The business lacks visibility into why.

Engineering the Solution: Project Timeline

Phase 1: Ingestion & Validation



Setting up Fabric Lakehouses, inferSchema CSV loading, and strict row-count preservation checks.

Phase 2: Transformation & Modelling



Complex PySpark deduplication, implementing the Haversine distance UDF, and overcoming session timeout challenges via Delta table persistence.

Phase 3: Visualisation & Analytics

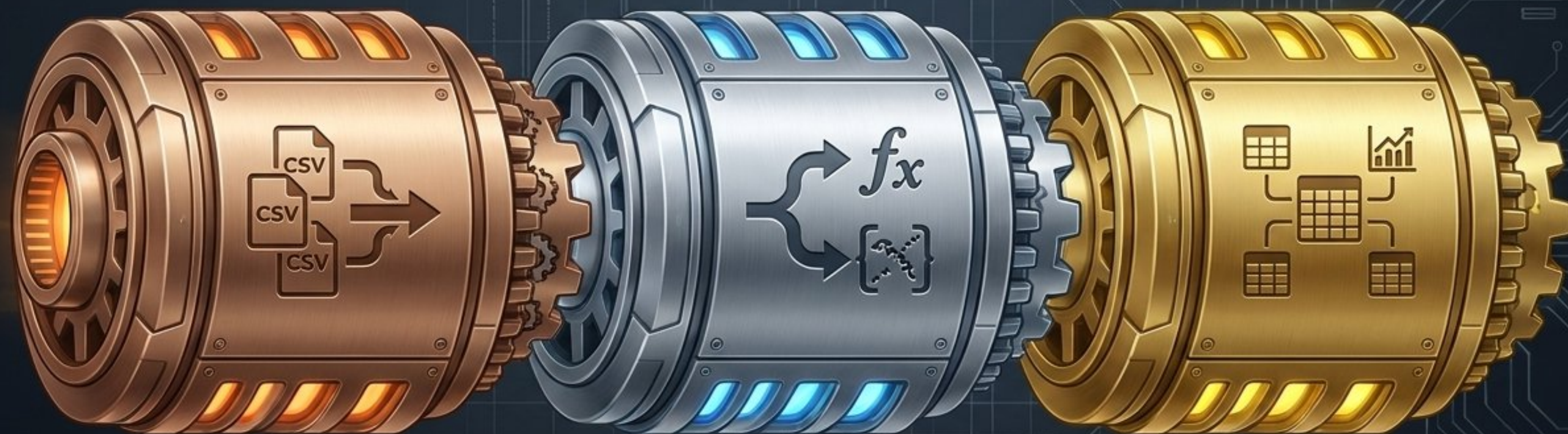


Designing the Gold Star Schema (handling Direct Lake DAX limitations natively in PySpark), Azure Maps integration, and final correlation testing.

Unified Analytics with Microsoft Fabric



The Medallion Data Engine



Bronze (Raw Ingestion)

Ingesting 9 raw CSV files untouched into OneLake using inferSchema to preserve data fidelity and track full history.

Silver (Cleansed & Enriched)

Complex deduplication, handling null values, and calculating the Haversine distance formula via Python UDF to map exact geographic coordinates.

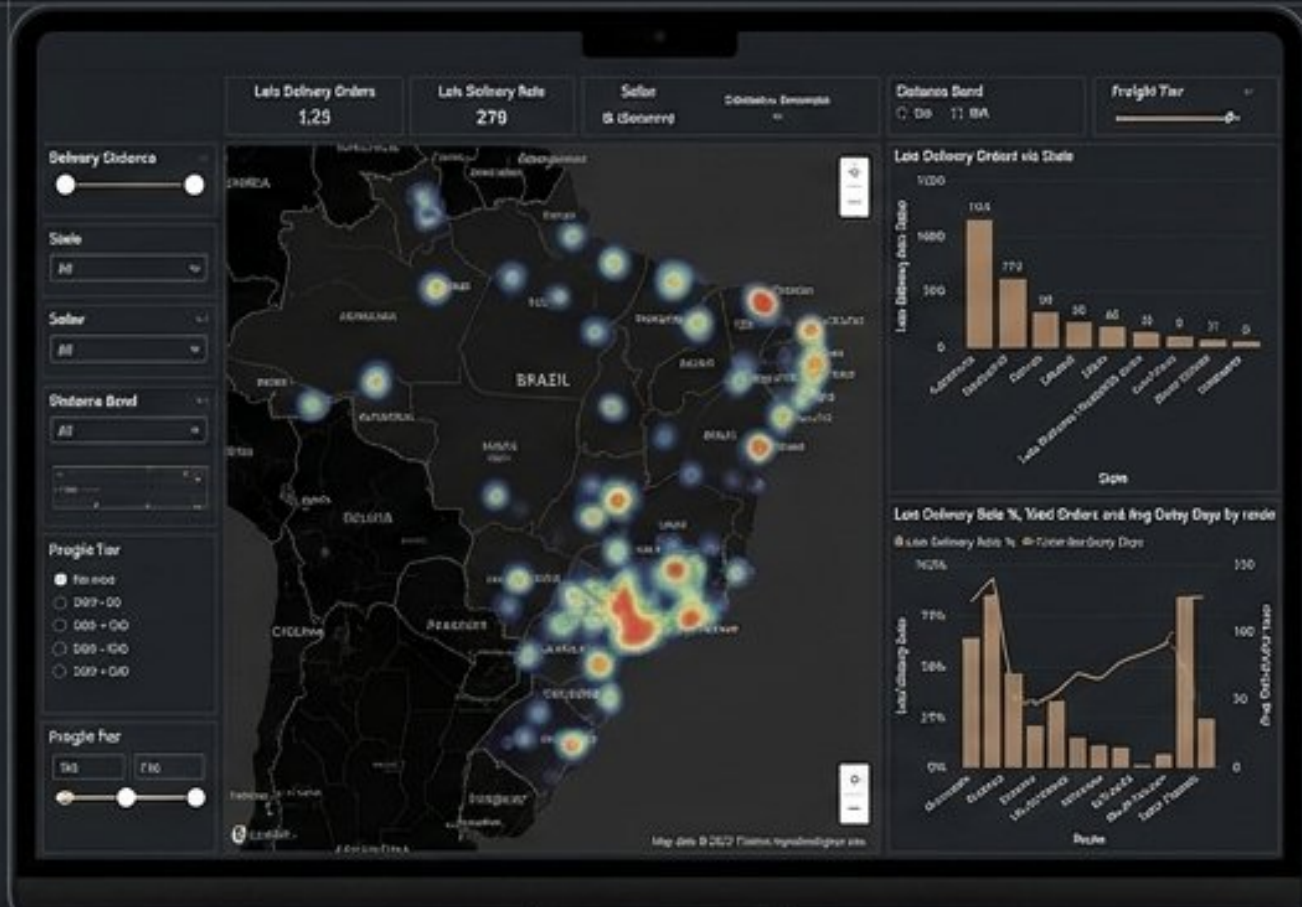
Gold (Business-Ready)

A highly optimised star schema engineered specifically for the Power BI engine and Direct Lake consumption.

Resolving Relational Complexity



From Lakehouse to Lighthouse



1

6-Page Interactive Dashboard

Powered by a Direct Lake connection for near-real-time querying with zero data duplication.

2

Geographic Hotspot Mapping

Bypassing standard state abbreviation errors by mapping exact Latitude/Longitude precision via Azure Maps.

3

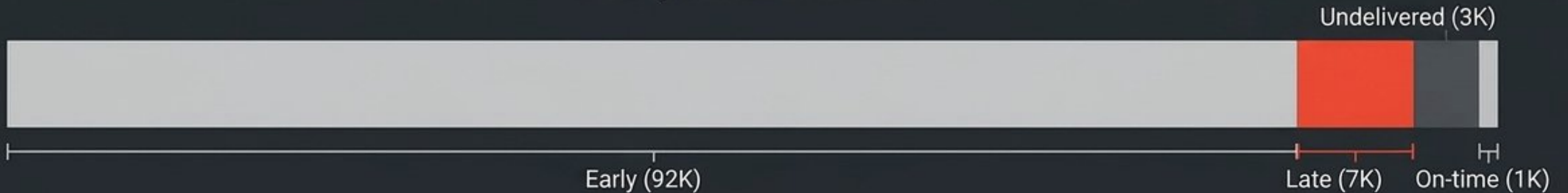
Self-Service Analytics

Empowering operations managers to filter instantly by state, seller, distance band, and freight tier.

Operational Baseline: The Scope of the Problem



Delivery Status Breakdown



While over 90% of deliveries succeed early, the **7,000+ late orders** experience extreme delays, pointing to systemic failures rather than minor operational friction.

Geographic Hotspots: The Regional Divide

High Volume, Manageable Risk

São Paulo (SP):
4.41% Late

Rio de Janeiro (RJ):
5.29% Late



Low Volume, Severe Risk

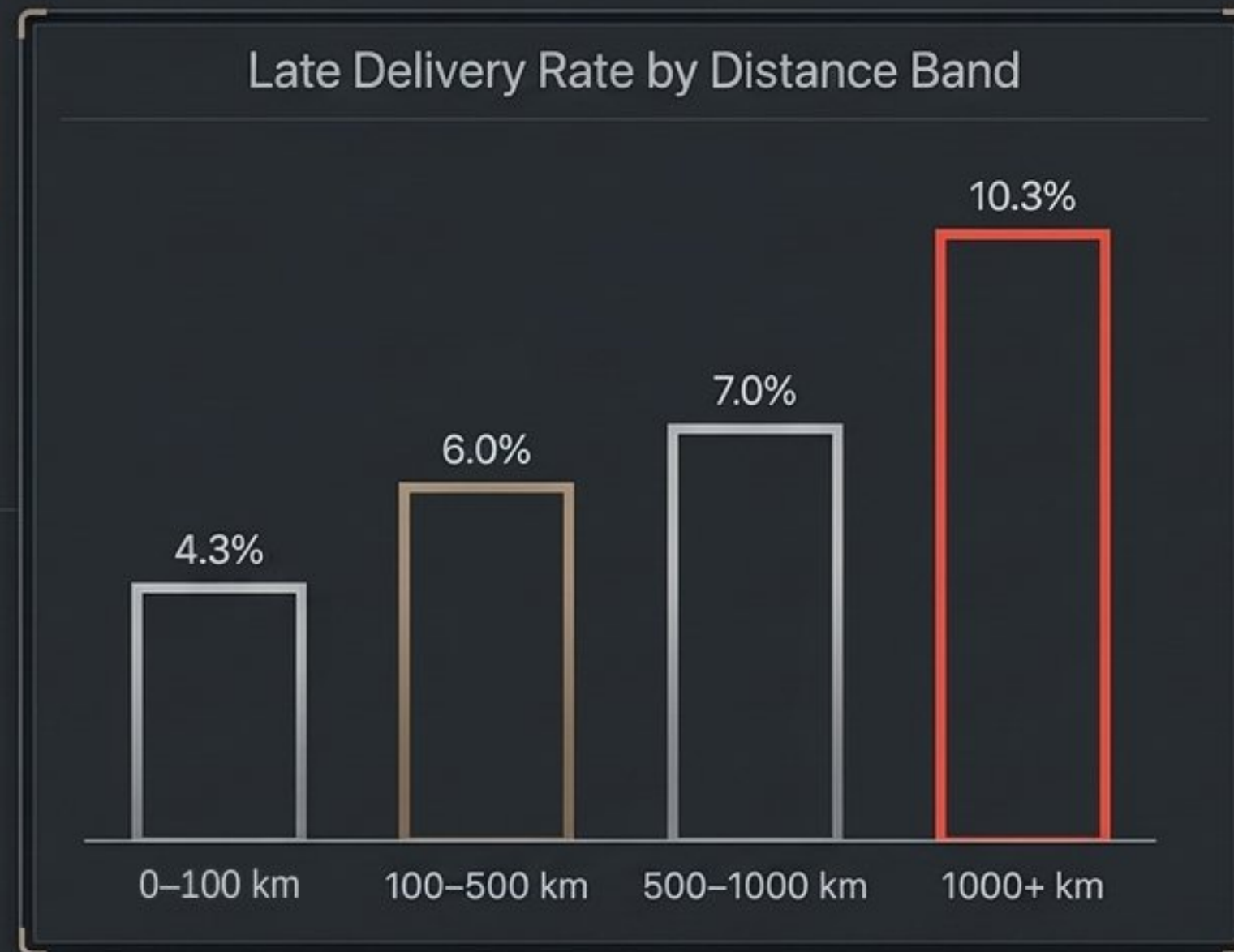
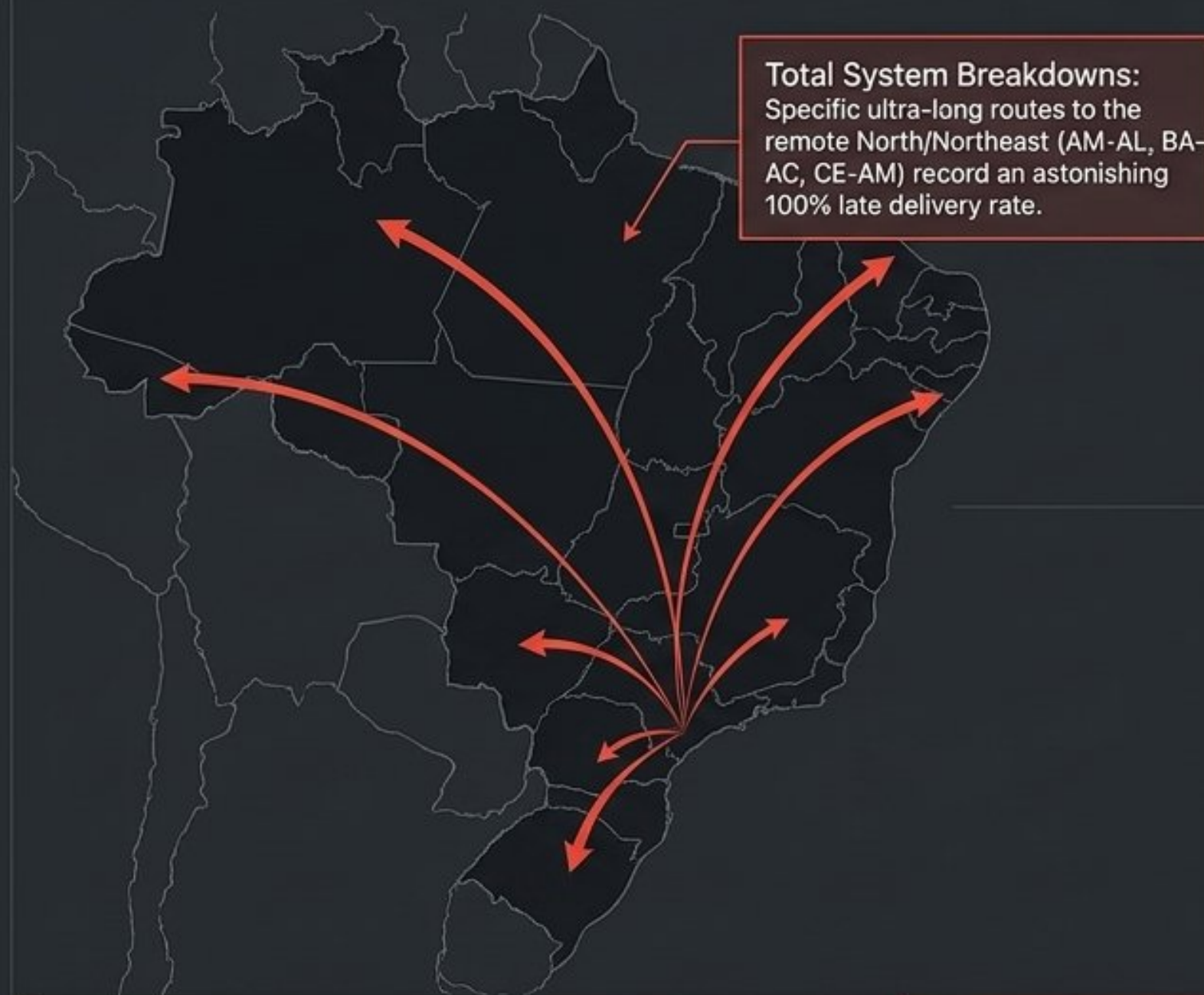
Espírito Santo (ES):
20.00% Late

Maranhão (MA):
18.18% Late

Intrastate Infrastructure Failure

Even short, within-state routes like ES–ES and MA–MA suffer extreme delays, suggesting localised logistical dead zones independent of pure distance.

Insight 1: The 1,000-Kilometre Threshold



Distance is the dominant causal factor of delay. The system is not designed to support 1000+ km last-mile logistics.

Insight 2: The Premium Freight Paradox

The Myth

Paying a higher freight cost guarantees faster delivery.

The Reality !

Average Freight for On-Time Orders:
R\$ 19.32

Average Freight for Late Orders:
R\$ 22.87

High-freight tier deliveries hit a 9.1% late rate (vs 4.5% for low freight).



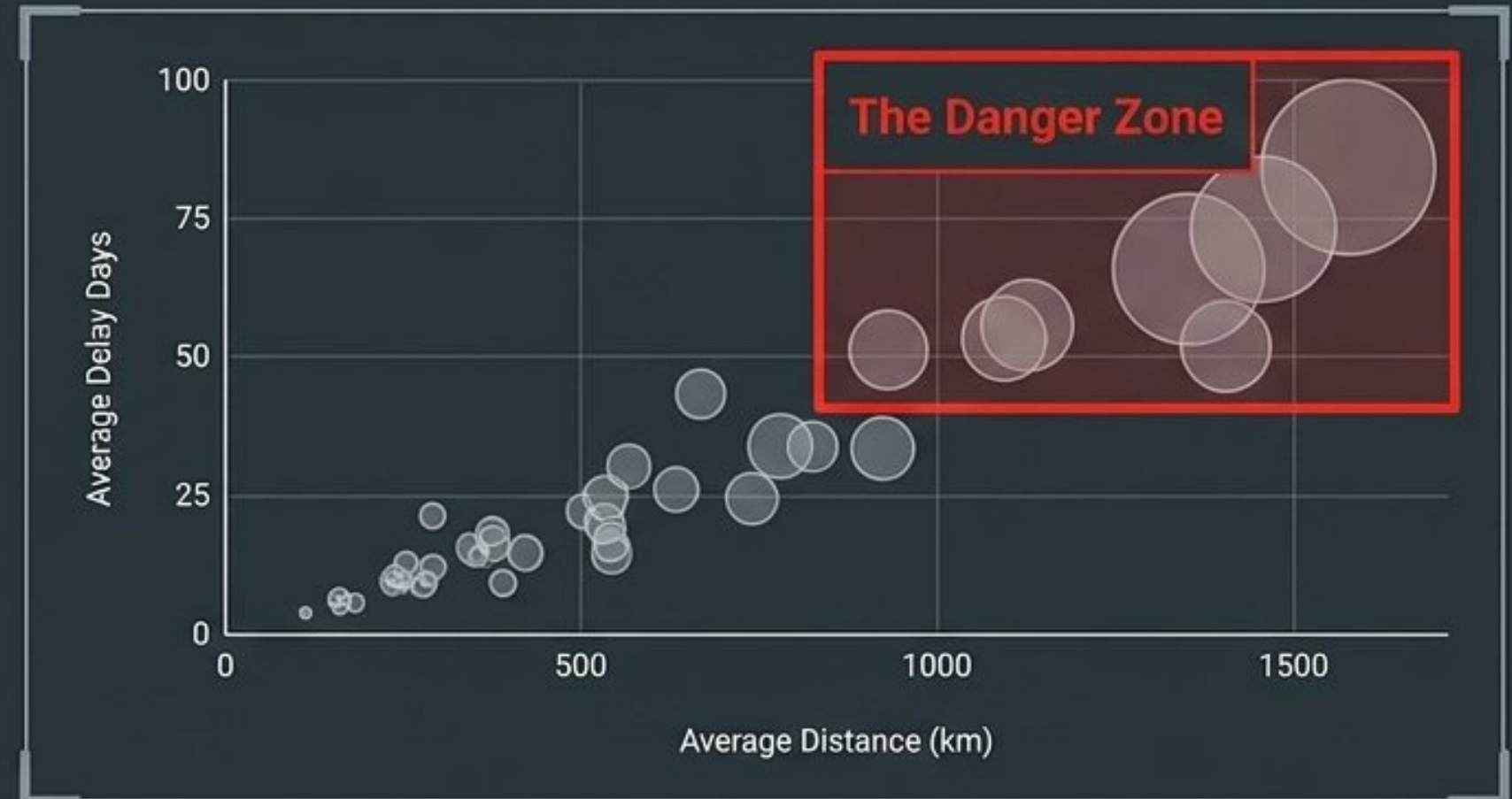
The Diagnostic Conclusion: Higher freight costs do not buy speed. Customers experiencing the worst service are paying a premium for it. Freight pricing reflects distance and physical size, not logistical quality.

Insight 3: The Compounding Effect of Weight

TACTICAL LOGISTICS BLUEPRINT

The Data

- Heavy parcels consistently cluster in states with the weakest infrastructure (AC, AM, MS, TO).

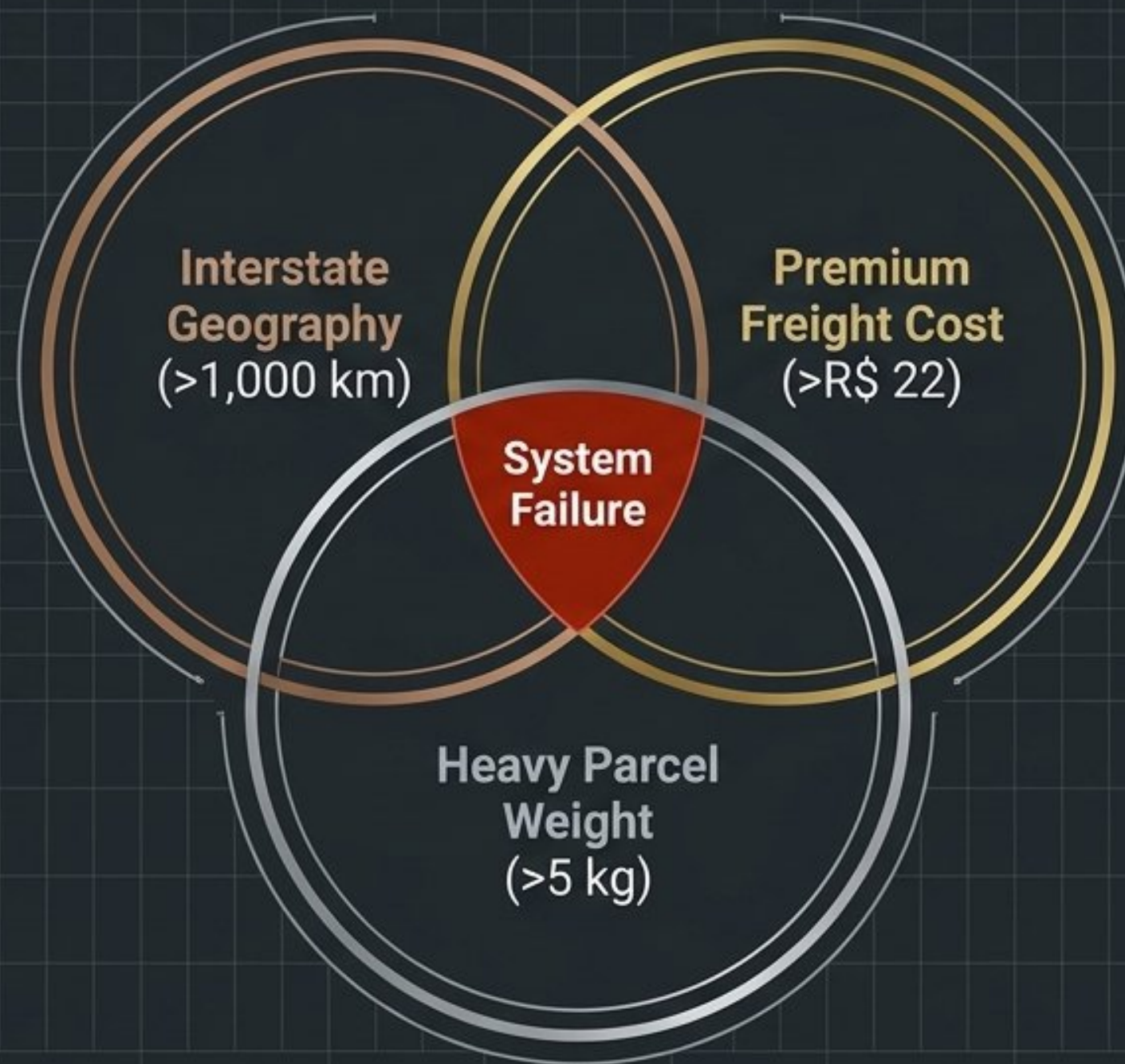


The Danger Zone: The data reveals a severe compounding effect. The combination of long-distance transport and heavy parcel weight creates the absolute highest-risk delivery profile within the Olist ecosystem.

Synthesis: The Anatomy of a Late Delivery

Delays are not random occurrences.

They are highly predictable intersections of geography, infrastructure limits, and product physics.



If Olist solves the logistics for heavy-parcel, long-distance interstate routes, the global late delivery rate will plummet.

Conclusion - Optimization Suggestions



Action 1: Regional Distribution

Establish regional cross-docking hubs in northern/northeastern states to eliminate the 1000+ km last-mile gap and bypass intrastate dead zones.



Action 2: Specialist Carriers

Partner exclusively with heavy-freight logistics specialists for all 5+ kg parcels destined for high-risk states (AC, AM, MS).



Action 3: SLA-Based Pricing

Realign the freight pricing strategy with performance-based carrier SLAs, ensuring premium pricing guarantees expedited service.

Olist Delivery Analytics



DAIH-Namic Four

Data Engineering Phase 1, 2 & 3 | Microsoft
Fabric + Power BI Integration

Optimise the Last Mile Today.

The Medallion data pipeline is built.
The insights are clear. The root
causes of failure have been mapped.

It is time to route the future of Olist.

Initiate Carrier Review &
Access Live Power BI Model